Omer Kilic

omer.kilic@rutgers.edu | (551) 775-3655 | North Jersey, NJ | linkedin.com/in/omer-kilic16 | omerkilicdev.github.io

Education

Rutgers, The State University of New Jersey — New Brunswick, NJ
B.S. Aerospace Engineering, GPA 4.00/4.00; Aresty Research Scholar; Peters Endowed Scholar — *Expected Graduation: May 2028*

Bergen Community College — Paramus, NJ
Associate of Science in Engineering Science, GPA 3.90/4.00; Phi Theta Kappa; U.S. Department of Agriculture 3SP Scholar — *2024 – 2025*

Experience

Aresty Research Assistant, Flapping-Wing Aerodynamics Lab — 2026 – Present
Rutgers Department of Mechanical and Aerospace Engineering; PI: Prof. Mitsunori Denda — *New Brunswick, NJ*

- Developed and verified a 3-D reduced-order wake model through an 80-step direct-versus-multipole comparison and 48-run sensitivity study, tracing late-time force error to far-wake deformation.
- Built a Julia verification library with 40 automated test groups and 274 checks and vectorized the exact aerodynamic kernel, reducing a representative simulation from 1,783 s to 50 s (~36× faster).

Engineering Intern (Candidate Engineer), Turkish Aerospace Industries (TUSAS) — Aug – Sep 2026
UAV Manufacturing & Fighter-Aircraft Aerodynamics; SKY Global Turk Internship Program (YTB) — *Ankara, Turkiye*

- Executed fighter-aircraft store-separation work from geometry preparation and CFD meshing through aerodynamic database generation and 6-DOF trajectory evaluation in MATLAB/Simulink.
- Mapped an 18-step composite aerostructure manufacturing flow from NX/Fibersim ply definition through layup, cure, machining, inspection, coating, assembly, and final acceptance.
- Evaluated a 3-stage rotor-blade balancing process using CG/moment relationships and gram-scale trim corrections, linking tooling, tolerance, coating, and mass variation to vibration and dynamic tracking.

Propulsion Engineer, AIAA Rutgers Rocket Propulsion Laboratory — Jan 2025 – Present
Solid Rocket Propulsion, Manufacturing & Test — *New Brunswick, NJ*

- Machined and fitted 6061-T6 motor hardware for a 98 mm solid rocket motor and hydrostatically proof-tested the assembled casing to 1,200 psi with no leakage.
- Led two static-fire tests producing ~5.4 and ~5.1 kN·s corrected total impulse; identified a thrust-channel zero offset accounting for ~8% of calculated impulse and used corrected data to refine propulsion models.

Electric Vehicle Conversion, Bergen Community College STEM Research Center — 2024 – 2025
Powertrain, Embedded Systems & Reliability — *Paramus, NJ*

- Reconfigured a 1998 Chevrolet K1500 EV from 35S1P to 35S2P, doubling stored energy from 20.7 to 41.4 kWh and theoretical current capability from 360 to 720 A while maintaining a 117 V pack.
- Led root-cause investigation across 4 vehicle test campaigns; after battery/BMS corrective actions, demonstrated ~450 A with no BMS shutdown and a 28°C peak on expansion-pack thermistors.
- Integrated an Arduino/CAN telemetry system for BMS, GPS, microSD, and live display; refactored embedded C++ by ~100 lines while correcting telemetry and logging defects.

Founder & Electrical Technician, Hi Tech Electrical Contractors Inc. — 2020 – Present
Electrical Systems & Field Troubleshooting — *Palisades Park, NJ*

- Founded and operate an electrical contracting business while studying engineering full time, leading projects from diagnosis and estimating through NEC-compliant installations, fault isolation, troubleshooting, and customer handoff.
- Grew the business to approximately $100,000 in first-year revenue and earned 22 verified five-star customer reviews as a Home Depot Pro Referral partner.

Research Intern, Laboratory for Materials Interfaces, NJIT — Summer 2025
PI: Prof. Gennady Gor — *Newark, NJ*

- Parameterized and executed 28 MezCal GCMC simulations, predicting 0.57–0.67 GPa He pressure and 0.13–0.16 mol/cm^3 density consistent with published nuclear-material ranges.

Technical Skills

Engineering Analysis: MATLAB/Simulink, Python, Julia, C++, numerical verification, sensitivity analysis, Monte Carlo methods, data reduction

Design & Manufacturing: SolidWorks, GD&T, ProtoTRAK lathe, machining, NX/Fibersim exposure, composites, Teamcenter exposure, NCR/MRB

Test & Systems: Static-fire testing, hydrostatic proof testing, CAN bus, Arduino, BMS diagnostics, instrumentation, sensor integration, electrical troubleshooting

Aerospace: CFD meshing/post-processing, 6-DOF dynamics, store separation, discrete-vortex methods, propulsion modeling, rotor/propeller balancing